SQL Database Project

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Topic:

UK Gaming Console Sales and Seasonal Promotion Analytics.

Project Overview:

In this project, I create a relational database by creating synthetic data for analysing UK gaming consoles sales and add a feature of seasonal promotions.

The dataset includes different models of gaming consoles over 2025 and part of 2026.

- **Sony PlayStation**: PS4, PS4 Slim, PS4 Pro, PS5 Fat, PS5 Slim and PS5 Pro.
- **Microsoft Xbox**: Xbox Series S and Xbox Series X.
- **Nintendo**: Nintendo Switch, Switch Lite, Switch OLED and Switch 2.

Realistic Approach:

1. The dataset contains 16 console models and 8 seasonal promotion records.
2. It generates 1,500 synthetic customers and 1,500 sales.
3. NULL values were introduced in the Promotion_ID column to simulate incomplete sales records and practise missing-value handling.
4. The Faker library was used to generate synthetic customer names, email addresses and UK-style postcodes.
5. Applied a 15% increase to the 2026 base price to simulate rising RAM costs and assess their impact on hardware pricing before discounts are applied.
6. Newer and more popular models are more likely to appear in sales, with different weights for 2025 and 2026.
7. Preventing Nintendo Switch 2 sales before its stated release date of 5 June 2025.
8. Creating seasonal promotions around were included in the dataset period.
    - New Year Sale.
    - Easter Sale.
    - Summer Sale.
    - Black Friday Sale.
    - Christmas Sale.
9. Applying discounts of 5%, 10%, 15%, or 20% only when a promotion is active; non-promotional sales retain a 0% discount.

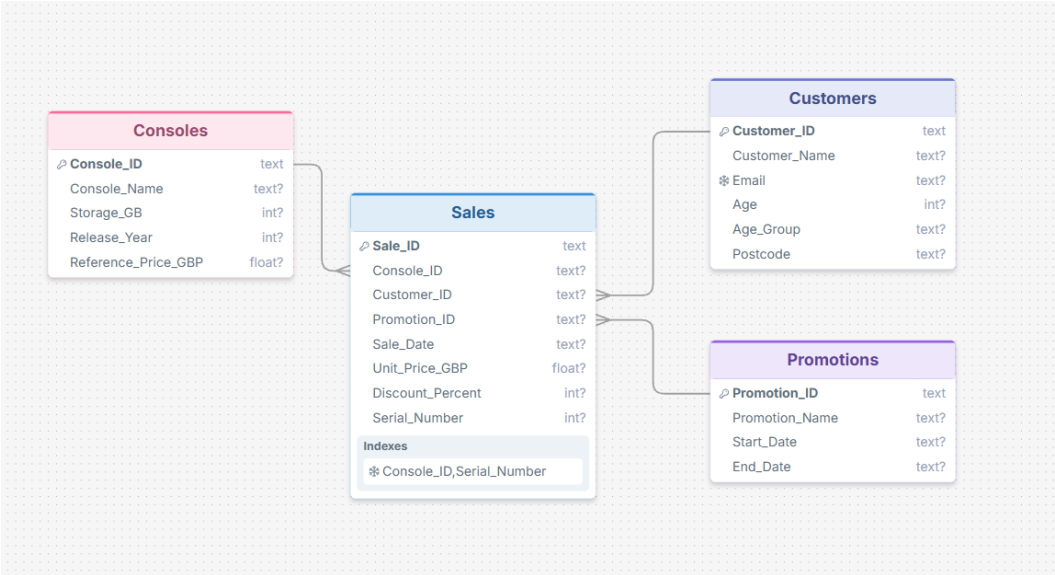
Tables and SQL Schema:

The database contains four main tables:

1. Consoles
2. Promotions
3. Customers
4. Sales

The database is divided into four tables to reduce unnecessary data repetition and improve data organisation.

- **Primary Keys (PK):** Console_ID, Promotion_ID, Customer_ID and Sale_ID uniquely identify records in their respective tables.
- **Foreign Keys (FK):** Console_ID, Customer_ID and Promotion_ID in the Sales table establish relationships with the other three tables.
- **Composite Unique Key:** Console_ID and Serial_Number together ensure that each console serial number is unique within its model.



Data Generation:

Table	Generation method
Consoles	Manually defined product catalogue.
Promotions	Manually defined seasonal promotions.
Customers	Faker and NumPy.
Sales	NumPy random generation, predefined rules and reference data.

This project demonstrates how Python and SQLite can be used together to create a structured relational database for gaming console sales analytics. Overall, the project provides practical experience in synthetic data generation, relational database design, data management and preparation for further data analysis.

Github Link:

https://github.com/kashannadeeem/Data-Mining-Assignment-1.git